

Assessment of Equatorial Plasma Bubble Impacts on Ground-Based Augmentation Systems in the Brazilian Region

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ABSTRACT

A preliminary Brazilian ionospheric anomaly threat model for Ground Based Augmentation Systems (GBASs) was developed from a comprehensive analysis using many days of ionospheric data from Brazil to reflect low-latitude conditions. Most anomalous ionospheric spatial gradients in this model were caused by nighttime Equatorial Plasma Bubbles (EPBs). In particular, the largest observed EPB-induced gradient (~ 860 mm/km) is two times larger than the upper bound (425 mm/km) on spatial gradients for the Conterminous U.S. (CONUS). The higher bound in the Brazilian ionospheric threat model has a significant effect on GBAS performance and availability. More detailed performance evaluation based on the Brazilian threat model is needed before GBAS can be made operational there.

This paper first assesses the performance of GBAS with Position Domain Geometry Screening (PDGS), which has been used in CONUS. Using the parameters of the preliminary EPB threat model, sets of worst-case geometries among an EPB gradient front and satellite Ionospheric Pierce Points (IPP) are generated. To calculate undetected range errors for hypothetical GBAS users, we derive an analytical ionosphere-induced differential range error model. Based on these inputs, we analyze the performance of GBAS under worst-case ionospheric conditions. The results show that inflating the standard deviation of the vertical ionospheric gradient only is not sufficient to completely remove potentially unsafe satellite geometries. This leads to additional inflation of the standard deviation of pseudorange correction errors. If the maximum EPB-induced gradient is larger than about 600 mm/km, the resulting CAtegory-I (CAT-I) GBAS availability is significantly degraded, making it difficult to meet the system requirement using this approach.

For this reason, we have also employed Monte Carlo analysis. The key difference from the previous approach

is that credit is taken for a prior probability of an extreme EPB event instead of assuming that worst-case storms occur with a probability of one in PDGS. This stochastic approach assesses the overall user impact by running many anomalous ionospheric trials based upon numerous combinations of threat model parameters. For each subset geometry, it is assumed that either one or two satellites are affected by the worst-case gradient while the remaining satellites are affected by anomalous but non-worst-gradients. Therefore, the worst-case impact should be approximated by inclusion in the distribution of simulation results. A real-time mitigation scheme based on the Monte Carlo method, which also requires sigma parameter inflation, is proposed that takes credit for the prior probability. System availability evaluation using the Monte Carlo approach with a single worst-case satellite impact shows that the availability penalty is significantly reduced compared to the previous (worst-case) method.

1.0. INTRODUCTION

Ground-Based Augmentation Systems (GBASs) that enhance the performance of Global Navigation Satellite Systems (GNSSs) must be robust to all possible ionospheric threats to avoid hazardous errors and frequent service interruptions. The Honeywell SLS-4000 GBAS ground facility installed at the Galeão International Airport (GIG) in Rio de Janeiro, Brazil was originally configured with the ionospheric threat model developed to cover the Conterminous U.S. (CONUS), which has a maximum spatial gradient of 425 mm/km [1]. However, equatorial ionosphere situated within 25 degrees latitude of the geomagnetic equator is known to be significantly more intense and unpredictable than what is encountered (and has been extensively studied) in mid-latitude regions such as CONUS. Therefore, models of ionospheric behavior developed for mid-latitude regions do not generally apply in equatorial regions. This motivated the development of a new threat model reflecting low-latitude

ionospheric conditions in Brazil based upon analyzing data collected in that region.

Assessment of Brazilian GBAS ionospheric spatial decorrelation was carried out by an international and interagency team comprised of experts on the ionosphere and GBAS starting from October 2013 [2]. This study utilized dual-frequency GNSS data collected from 2011 to 2014 to assess Brazilian ionospheric behavior. Over 1000 severe ionospheric gradients were identified from post-processed data, and almost all of these points were induced by nighttime Equatorial Plasma Bubbles (EPBs). Including the largest observed slant gradient of about 850 mm/km, a significant number of events exceeded the upper bounds (375-425 mm/km) of the CONUS threat model [1].

In addition to gradient, other threat model parameters describing the geometry of EPBs, including occurrence time, zonal length, transition zone length, maximum depletion, propagation speed, and tilt angle, were defined and estimated [2]. Preliminary parameter estimates were made from the EPBs which produced slant gradients larger than 400 mm/km. The results showed that EPBs occurred almost always during post-sunset hours between 1800 and 0500 local time, with the peak in the late evening just before local midnight. Their zonal scales (edge-to-edge distances) varied from 96 km to 750 km. A maximum depletion of 35 meters and transition zone lengths of 20-450 km were estimated. These EPBs moved roughly eastward with speeds of about 40 to 250 m/s and were slanted with tilt angles of within ± 35 degrees measured from geomagnetic north. This preliminary Brazilian ionospheric threat model should be incorporated into EPB impact simulations to assess differential ionospheric errors and their impact on GBAS users in the Brazilian region.

To assess EPB impacts on CAT-I GBAS users, this paper implements two methods; "Position Domain Geometry Screening" (PDGS) and "Monte Carlo analysis". The well-known method of PDGS defined in [5] is modified to simulate and evaluate EPB impact scenarios, whereas the Monte Carlo method is newly developed. Section 2.0 derives an analytical ionosphere-induced range error model based on a mathematical approach. Section 3.0 and 4.0 present the ionospheric impact scenarios in the presence of EPBs and compare GBAS availability results obtained from the two methods. In Section 5.0, we draw conclusions along with some thoughts on future work.

2.0. ANALYTICAL MODELING OF IONOSPHERE-INDUCED RANGE ERROR

This section describes how we model differential errors in range domain caused by ionospheric front impacts. The Code Carrier Divergence (CCD) monitor within the GBAS ground facility is capable of detecting large ionospheric gradients by measuring the time rate of change of divergence between code and carrier

measurements. The detection capability of the CCD monitor relies on a relative speed of the ionospheric front to the velocity of a satellite IPP projected onto a line perpendicular to the front ($\Delta v = |v_{IPP,proj} - v_{front}|$). As Δv is small, the ionospheric delay change in time observed by the ground facility is not fast enough to be detected by the CCD monitor. This may result in unacceptably large residual errors in GNSS range signals, even after applying differential corrections.

Undetected ionosphere-induced differential range errors can be calculated for hypothetical GBAS users using ionospheric gradient magnitudes, combined with the worst-case relative speed of an ionospheric front with respect to a satellite IPP, Δv . A closed-form expression for the ionosphere-induced range error (as a function of Δv) was originally suggested in [3]. This was used in the worst-case ionospheric anomaly impact simulation for CONUS and is still useful. However, the uncertainty in the approximated error model cannot be ignored as the magnitude of gradient becomes larger, as it does in the Brazilian threat model. This motivates the derivation of an improved closed-form expression, which was derived based on Laplace transform analysis and a search for the worst-case range domain errors within the GBAS service volume. With two input variables (Δv and gradient magnitude), the resulting range error model enables us to conservatively cover all possible range errors under worst-case ionospheric impact scenarios.

2.1 Exhaustive search for ionosphere-induced differential range error

Code-carrier divergence caused by ionospheric gradients induces bias errors into the process of code-carrier smoothing. These errors in Carrier-Smoothed Code (CSC) can be amplified as either a smoothing time or a magnitude of gradient becomes larger. In GBAS, both aircraft and the ground facility use a 100-sec time constant for the smoothing process. The resulting differential range errors can be computed by taking the difference between the smoothed ionospheric delay measurements observed from aircraft and the ground facility. We derived mathematical expressions of the CSC errors and the CCD monitor outputs based on Laplace transform analysis.

For CAT-I GBAS, the CCD monitor uses a 30-sec filter time constant [4]. Instead of using the threshold defined in [4], we take a more conservative approach by using the Minimum Detectable Error (MDE) of 0.04 m/s as a threshold for the CCD monitor. For a set of simulation parameters (gradient, width, Δv , initial positions of ionospheric front and aircraft), the differential range errors are simulated over time as the ionospheric front and aircraft move toward the ground facility. Then, we compute and record the range error either detected or undetected by the CCD monitor when the aircraft arrives at CAT-I DH. Using the exhaustive simulation algorithm

[5], we search the maximum differential range errors over all possible combinations of these parameters.

2.2 Linear model for closed-form error expression

As a result of the exhaustive search, the differential range error curve can be expressed as a function of Δv and gradient magnitude. Figure 1 shows the differential range errors (blue dashed curve) as a function of Δv when a front with the worst gradient of 860 mm/km (shown in [2]) is assumed to hit aircraft. A simple linear model of closed-form expression (red line) is proposed to conservatively bound the simulated range error curve. The linear model can be expressed as Eq. (1)

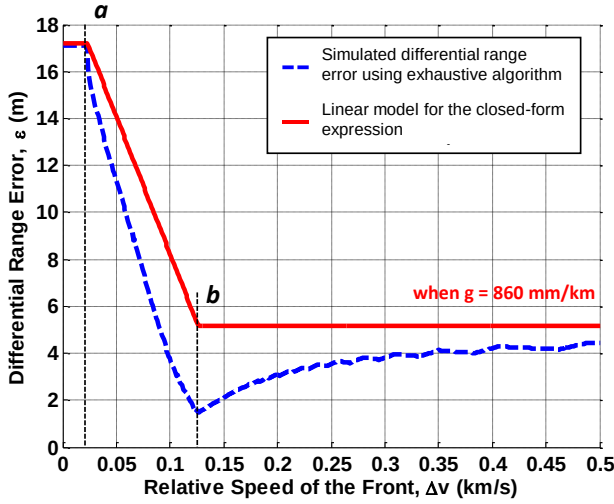


Figure 1. Example of modeling of ionosphere-induced errors ($g=860\text{mm/km}$): Differential range errors simulated using the exhaustive algorithm (blue dashed curve) and linear model for the closed-form expression (red solid curve).

$$\begin{aligned} \varepsilon &= \min\left(\frac{35}{w}, g\right)(x_{\text{air}} + 2\tau v_{\text{air}}) \quad \text{if } \Delta v \leq a \\ \varepsilon &= \frac{2\tau v_{\text{air}} g}{a - b}(\Delta v - b) + gx_{\text{air}} \quad \text{if } a < \Delta v \leq b \\ \varepsilon &= gx_{\text{air}} \quad \text{if } b < \Delta v \end{aligned} \quad (1)$$

$$a = \frac{0.02}{\min\left(\frac{35}{w}, g\right)}, \quad b = \frac{0.014}{g} + 0.11$$

where w represents a width of the ionospheric front, τ is a 100-sec time constant of single-frequency GBAS code-carrier-smoothing filter, and v_{air} is the speed of aircraft (70 m/s in this paper). Δv refers to the relative speed of the ionospheric front with respect to a satellite IPP. a is a point of Δv at which the CCD monitor starts to detect

the anomalous ionospheric gradient, and b is a point of Δv at which the differential range error is at a minimum.

The closed-form expressions are divided into three regions as a function of the relative speed, Δv . When $\Delta v \leq a$, the CCD monitor cannot detect anomalous ionospheric gradients, as explained above. Without the CCD monitor's tripping, ionosphere-induced range errors are proportional to the magnitude of gradient (g) and the effective separation between aircraft and the ground facility. The effective separation is defined as the sum of a physical separation (x_{air}) between the aircraft and the ground facility and a synthetic separation ($2\tau v_{\text{air}}$) due to the memory effect of the single-frequency carrier-smoothing filter.

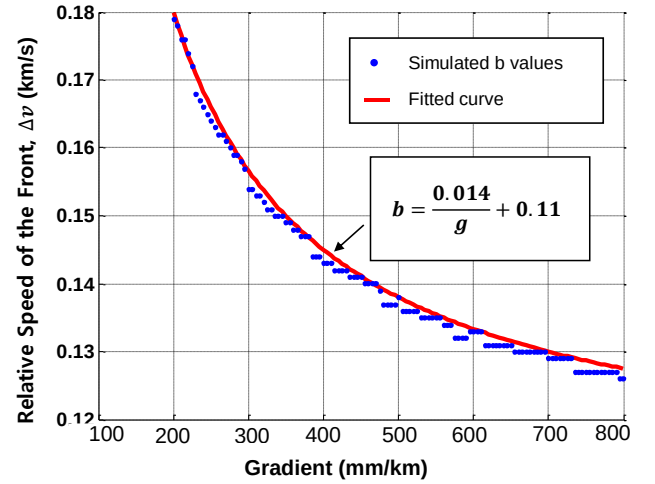


Figure 2. b values as a function of gradient.

When $a < \Delta v \leq b$, the CCD monitor is capable of detecting anomalous ionospheric gradients and the resulting differential range error decreases to the minimum error at b as Δv increases. The effect of ionospheric dispersive media on code and carrier measurements have the same magnitude but opposite sign. Since carrier measurements dominate in the carrier smoothing filter directly after the ionospheric front impact, the initial response of the smoothed code error goes into the opposite direction. The CSC errors in the negative direction result in differential range errors which are smaller than the errors which depend on only the physical separation (gx_{air}). At a certain relative speed point (b), the effect of negative errors in the CSC measurements is maximized, and a transition of the error state takes place. We used this point b for the linear error model. The b values can be modeled as a function of gradient as shown in Figure 2.

When $b < \Delta v$, the CCD monitor trips quickly and the memory effect of the CSC is minimized. The differential error converges to gx_{air} .

3.0. ASSESSMENT APPROACH I: WORST-CASE GRADIENT SIMULATION AND POSITION DOMAIN GEOMETRY SCREENING

This section describes the overall procedure to analyze the performance of CAT-I GBAS in the Brazilian region in the presence of EPBs using the current worst-case simulation and mitigation method. The well-known method of Position Domain Geometry Screening (PDGS) has already been used in CONUS [6]. This method simulates the worst-case undetected vertical ionospheric errors assuming the worst case SV-user geometry and parameter combinations. Note that this process lays the worst case parameters on top of each other and does not consider any probabilistic averaging among threat model parameter combinations.

Ionospheric observations from Brazil like those presented in [2] demonstrate that EPB characteristics are very much different from those of ionospheric storms in mid-latitude regions. Thus, we modified the worst-case impact scenarios (i.e., previous CONUS procedure) based on the characteristics of EPBs analyzed in [2]. The revised model of ionospheric induced range errors presented in Section 2.0 was also applied for the worst-case impact simulation. The assessment of EPB impact on GBAS consists of three steps: calculation of the ionosphere-induced position errors under worst-case ionospheric scenarios, determination of the broadcast sigma inflation factors for mitigating EPB-induced threats, and an analysis of the resulting GBAS performance.

3.1 Calculation of the ionosphere-induced position errors under the worst-case ionospheric scenarios

The analytical model of ionosphere-induced differential range error was introduced in Section 2. Using this model, we examine the largest potential errors resulting from the worst case scenarios. In this paper, we propose two sets of worst-case geometries among EPB and satellite Ionospheric Pierce Points (IPPs): the worst double-satellite impact scenario and the worst single-satellite impact scenario.

The double-satellite impact scenario considers two satellites affected simultaneously by the same front where an EPB transition zone starts. This scenario is organized in a similar manner to that of the CONUS threat simulation. The selection of satellite pairs is specified within the orientation angle constraint given by the limit on the EPB tilt angle (i.e., within ± 35 degrees from geomagnetic north). The double satellite impact scenario is repeated for all possible pairs of satellites of a particular subset geometry.

For a given pair of satellites (k_1 and k_2), IPP velocities projected onto the line perpendicular to the EPB front ($v_{k_1,proj}$ and $v_{k_2,proj}$) are calculated. Then, four sets of Δv_{k_1} and Δv_{k_2} are determined using the following logic. If $v_{EPB\min} \leq v_{k_1,proj} \leq v_{EPB\max}$, $v_{k_1,proj}$ is set to be the same as the velocity of the EPB front, v_{EPB} . Then, Δv_{k_1} is zero and $\Delta v_{k_2} = |v_{k_2,proj} - v_{k_1,proj}|$. Note that $v_{EPB\min}$ and $v_{EPB\max}$ are 40 and 250 m/s, respectively. If $v_{k_1,proj}$ is not within the EPB velocity bounds, this case is discarded. The same process is conducted for $v_{k_2,proj}$. The third set is determined by subtracting $v_{k_1,proj}$ and $v_{k_2,proj}$ from the lower bound of the EPB velocity, $v_{EPB\min}$, and taking absolute values of the results. For the last set, we perform a similar process to the third one but for $v_{EPB\max}$ (i.e., $\Delta v_{k_1} = |v_{k_1,proj} - v_{EPB\max}|$ and $\Delta v_{k_2} = |v_{k_2,proj} - v_{EPB\max}|$).

For the given worst-case gradient and the relative speed of the front with respect to satellite IPPs, Δv_{k_1} and Δv_{k_2} , each set of ionospheric range errors (ε_{k_1} and ε_{k_2}) can be obtained from the range error model in Eq. (1). Ionosphere-induced Errors in Vertical domain (IEVs) are calculated for each Δv set using Eq. (2)

$$\begin{aligned} IEV_{k_1k_2} &= |S_{vert,k_1}\varepsilon_{k_1}| + |S_{vert,k_2}\varepsilon_{k_2}| \\ S &= (G^T \cdot W \cdot G)^{-1} \cdot G^T \cdot W^{-1} \\ G_i &= [-\cos El_i \cos Az_i \quad -\cos El_i \sin Az_i \quad -\sin El_i \quad 1] \\ W^{-1} &= \begin{bmatrix} \sigma_1^2 & 0 & \dots & 0 \\ 0 & \sigma_2^2 & 0 & \dots \\ \dots & 0 & \dots & 0 \\ 0 & \dots & 0 & \sigma_M^2 \end{bmatrix} \\ \sigma_i^2 &= \sigma_{pn_gnd,i}^2 + \sigma_{tropo,i}^2 + \sigma_{air,i}^2 + \sigma_{iono,i}^2 \end{aligned} \quad (2)$$

where $S_{vert,i}$ is the vertical position component of the weighted-least squares projection matrix S for satellite i (refer to Sec. 2.3.10.2 in [7]), ε_i is the ionosphere-induced range error for satellite i , G_i is the i th row of the observation matrix G , corresponding to the i th satellite in view (satellite i , for simplicity). El_i is the elevation of satellite i , Az_i is the azimuth of satellite i , W^{-1} is the inverse of the least squares weighting matrix, and σ_i^2 is the variance of a normal distribution that overbounds the true post-correction range-domain error distribution for satellite i under the fault-free hypothesis.

Satellite pairs whose orientation angle is beyond the feasible range of EPB tilt angle are not selected by the double satellite impact scenario. In this case, the satellites are individually simulated as single-satellite impact scenarios. Assuming that EPBs move eastward along the geomagnetic equator, the propagation direction of the EPB is determined at the given latitude and longitude of satellite IPP (k_1). We then project the velocity of the satellite IPP onto the EPB propagation direction ($v_{k_1,proj}$), and compare the projected velocity to the feasible range of EPB velocity. If $v_{EPBmin} \leq v_{k_1,proj} \leq v_{EPBmax}$, Δv_{k_1} is set to zero. Otherwise,

$\Delta v_{k_1} = \min(|v_{k_1,proj} - v_{EPBmin}|, |v_{k_1,proj} - v_{EPBmax}|)$. Using Δv_{k_1} , IEV for a single satellite k_1 can be obtained from Eq. (3).

$$IEV_{k_1} = |S_{vert,k_1} \epsilon_{k_1}| \quad (3)$$

After calculating IEVs for both scenarios, the maximum IEV (MIEV) is selected for a particular subset geometry. In this paper, we assume that up to two satellites from the all-in-view satellites may become unusable due to ionospheric scintillation and final approach operations. Thus, we simulate the MIEVs over all-in-view, 1-SV-out, and 2-SV-out geometries.

3.2 Determination of broadcast sigma inflation factors for mitigating EPB-induced threats

Potentially hazardous subset geometries are determined by comparing MIEV of each subset geometry to the CAT-I Tolerable Error Limit (TEL) in the vertical domain (28.78 m at the minimum decision height of 200 ft) [8]. Every subset geometry whose MIEV exceeds TEL must not be used by GBAS users. The key for mitigating potential threats is to selectively increase the broadcast integrity parameters and consequentially elevate Vertical Protection Levels (VPLs) above the Vertical Alert Limit (VAL) [6]. VPL is calculated at each epoch as shown in Eq. (4)

$$VPL_{H0} = K_{ffmd} \sqrt{\sum_{i=1}^M S_{vert,i}^2 \sigma_i^2}$$

$$VPL_{eph} = \max(VPL_{eph,k}) \quad (4)$$

$$VPL_{eph,k} = |S_{vert,k}| x_{air} P_k + K_{md_{eph}} \sqrt{\sum_{i=1}^N S_{vert,i}^2 \sigma_i^2}$$

where VPL_{H0} is the vertical protection level under the fault-free hypothesis. K_{ffmd} is the multiplier (unitless) that determines the probability of fault-free missed detection.

A K_{ffmd} of 5.81 is used assuming that four ground subsystem reference receivers are installed. VPL_{eph} is the vertical protection level under a single-satellite (satellite k) ephemeris fault, and P_k is the ephemeris error decorrelation parameter for satellite k ($P_k = 0.00018$ is used in this paper; refer to [9],[10]). $K_{md_{eph}}$ is the multiplier (unitless) derived from the probability of missed detection given that there is an ephemeris error in a GPS satellite ($K_{md_{eph}} = 3.80$ is used in this work).

In this paper, we choose to inflate σ_{vig} and σ_{pr_gnd} . We first inflate σ_{vig} until it reaches the maximum allowable value of 25.5 mm/km and then inflate σ_{pr_gnd} as needed.

σ_{vig} is the standard deviation of a normal distribution associated with residual ionospheric uncertainty due to nominal spatial decorrelation ('vig' stands for vertical ionospheric gradients). $\sigma_{pr_gnd,i}$ is the total fault-free one-sigma ground error term associated with the corresponding differential correction error for satellite i .

3.3 Results from EPB impact assessment

A ground facility was assumed to be located at the Galeão International Airport (GIG) in Rio de Janeiro where the Honeywell SLS-4000 GBAS ground facility was installed. Two distinct simulations for the daytime (06:00 – 18:00 LT) and the nighttime (18:00 – 06:00 LT) are conducted separately with two different σ_{vig} values (9 mm/km in daytime and 15 mm/km in nighttime). During the daytime, we used the CONUS gradient bounds of 375-425 mm/km. In contrast, during the nighttime, we conducted multiple simulations by varying the constant gradient bounds from 500 up to 860 mm/km without elevation dependence.

The simulations were conducted for the Standard RTCA 24 satellite GPS constellation [11]. We simulated the worst-case ionosphere-induced errors at distances of 0 to 7 km from Decision Heights (DHs). DH is assumed to be located at 1 to 4 km so that the total 32 (8 times 4) cases are simulated at each epoch. Using the PDGS method, we identified potentially unsafe geometries and generated the inflation factors for broadcast sigmas which make them unavailable.

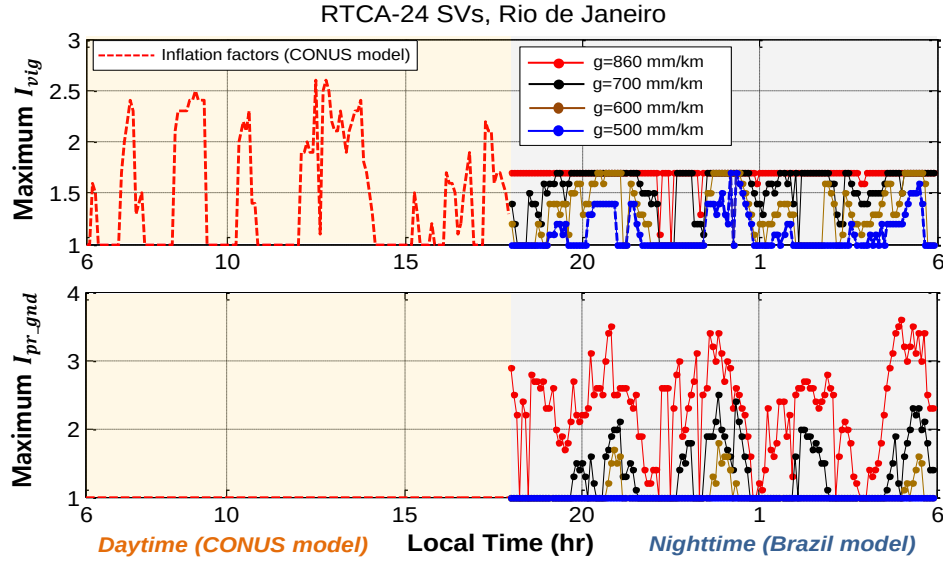


Figure 3. Maximum inflation factors of (a) σ_{vig} and (b) σ_{pr_gnd} among 32 cases at each epoch. Different colors indicate different maximum gradient bounds.

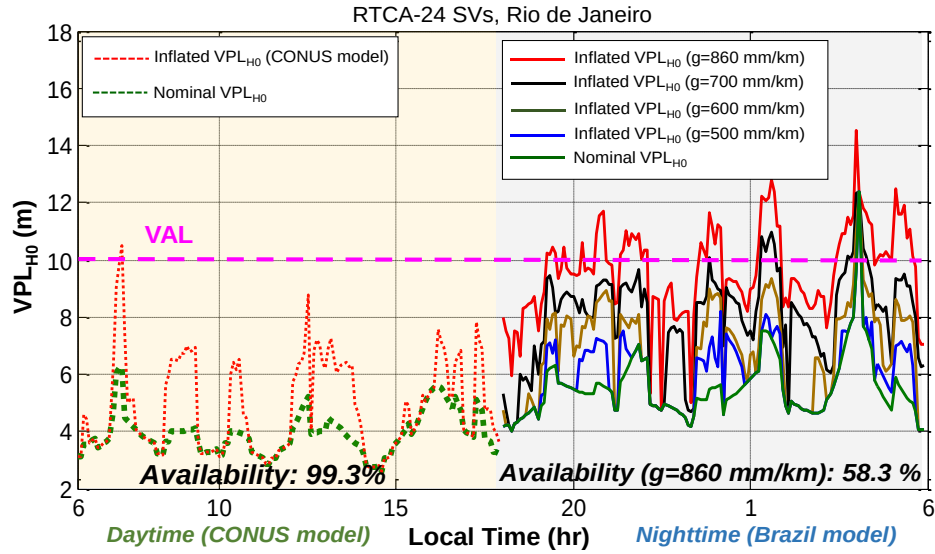


Figure 4. Inflated VPL_{H0} , nominal VPL_{H0} , and VAL at 4-km DH using the PDGS mitigation method

Figure 3 shows the maximum inflation factors among 32 cases at each epoch. During the daytime, σ_{pr_gnd} is not required to be inflated because the inflation factor of σ_{vig} at each epoch does not reach its maximum allowable limit, 2.8. However, during the nighttime, inflating σ_{vig} only is not sufficient to completely remove potentially unsafe geometries when the EPB-induced ionospheric gradient exceeds 600 mm/km. Note that the maximum allowable limit on the inflation factor of σ_{vig} is 1.7 during the nighttime since a σ_{vig} of 15 mm/km larger than that during the daytime was used. This leads to an additional inflation of σ_{pr_gnd} .

By applying these inflation factors in VPL computation, we assess the system availability under the worst-case ionospheric conditions. Figure 4 shows the inflated VPL_{H0} at 4-km DH during the daytime and the nighttime. The VPL_{H0} curves in Figure 4 were derived for all-in-view satellite geometries. The green curves indicate the nominal VPL_{H0} before inflating broadcast sigma. The different colors of the solid curves indicate the different size of the maximum gradient bound applied to EPB impact simulations. During the daytime, the inflation of σ_{vig} results in a slight loss on the availability (99.3%). In contrast, as the maximum gradient bound increases during the nighttime, the margin between VPL and VAL become smaller due to higher sigma inflation required. In

particular, when the gradient bound is 860 mm/km, the inflated VPL_{H0} (red solid curve) significantly exceeds VAL and the resulting availability decreased to 58.3%. Even with a gradient bound of 600 mm/km, the system undergoes non-negligible loss of availability, and the estimated availability is 98.61 %. These results imply that with an elevated gradient bound greater than about 600 mm/km, it is difficult to meet the CAT-I availability requirement using the current PDGS method.

4.0. ASSESSMENT APPROACH II: MONTE CARLO SIMULATION

Unlike the worst-case impact simulation, Monte Carlo simulation enables us to assess the overall (ensemble) user impact by running many anomalous ionospheric trials through the use of randomly generated parameters within the bounds of the threat model.. It allows us to combine worst-case impacts on individual satellites with random (non-worst-case) conditions on others to provide more-realistic combinations of range (and thus position) domain errors.

4.1 Monte Carlo simulation procedure

As shown in Figure 5, for each epoch and specific satellite geometry (comprised of M satellites), we randomly sample the EPB threat model geometry parameters, including tilt angle, width, and velocity, from uniform distributions. The lower and upper bounds of the uniform distribution for each parameter are given by the preliminary EPB threat model [2]. With a given EPB front direction specified by the randomly sampled tilt angle, satellite IPP velocities projected onto the direction of motion of the EPB are computed.

In these simulations, we further assume that either one or two satellites are affected by the worst case EPB with the max gradient of 860 mm/km. Thus, we select one worst-case satellite or two worst-case satellites in a given subset geometry and apply the maximum gradient. The Δv values for these two cases are determined in the same manner as for PDGS.

On the other hand, it is hypothesized that the remaining satellites are affected by anomalous but non-worst-case gradients. These gradients are not as extreme as the worst-case gradients but are still severe enough to induce large ionospheric errors for GBAS users. These non-worst-case gradients are assumed to be characterized by a Gaussian distribution derived from empirical gradient data (excluding the days when extreme gradients were observed). The set of Δv values is determined by taking absolute differences between the projected satellite IPP velocities and the EPB propagation velocity, which is randomly sampled.

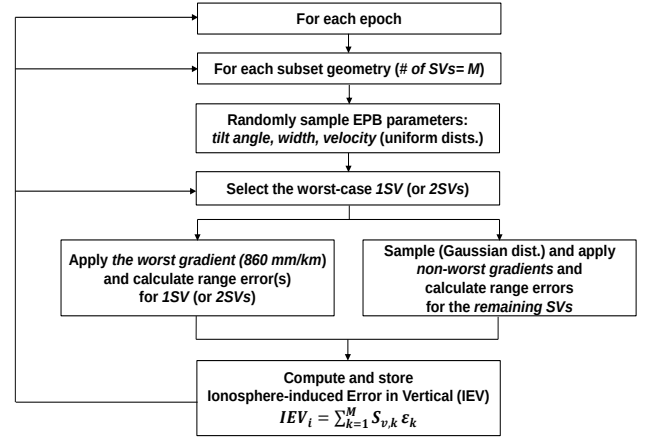


Figure 5. Monte Carlo simulation procedure

With the set of Δv values and the selected gradients for a specific subset geometry, we obtain ionosphere-induced differential range errors based on the model established in Section 3.0. Then, we compute and store ionospheric error in vertical domain (or IEV) for a particular subset (comprised of M satellites) using Eq. (5). Since every satellite could be affected by the worst-case EPB, we assign one satellite (or two) as the worst-case satellite (s) and then loop through all possible satellite combinations, as shown in Figure 5.

$$IEV = \sum_{i=1}^M S_{vert,i} \epsilon_i \quad (5)$$

In this procedure, an appropriate random sampling strategy over subset geometries is required because, if we do naïve sampling the sample set would be dominated by unlikely subset geometries. As an example, suppose that there are M satellites in view. Possible subset geometries are the single ${}_M C_0$ all-in-view geometry, ${}_M C_1$ 1-SV-out geometries, and ${}_M C_2$ 2-SV-out geometries. Although it is most probable that the aircraft uses all-in-view geometry for positioning, the simulated errors will be dominated by the 2-SV-out geometries which have the largest number of permutations. Instead, in this paper, we arrange the sampling to obtain 80% of the total samples from the all-in-view geometry, 15% from the set of 1-SV-out geometries, and 5% from the set of 2-SV-out geometries on each epoch. This weighting is chosen arbitrarily based on our estimate to generate a more realistic distribution of sampled position errors.

We repeat this procedure tens of thousands times to obtain a sample size of 10^7 for each time epoch. After simulating all possible geometries at each epoch, we examine the probability of distribution of IEVs with respect to TEL bounding. Because of this large number of samples, the actual “worst-case” scenario should be approximately included in the distribution of the simulation results.

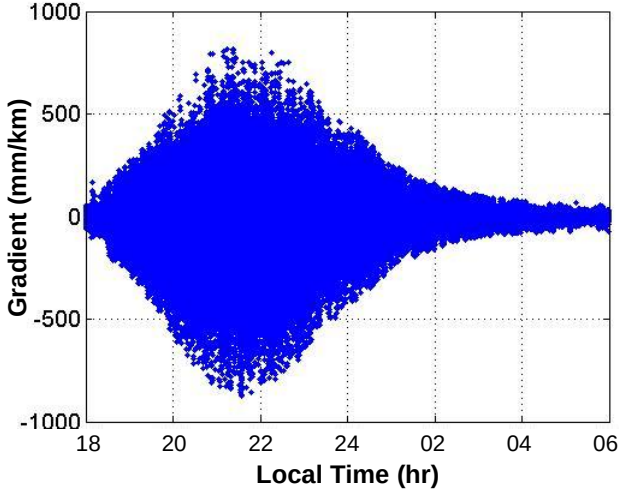


Figure 6. Ionospheric spatial gradients computed using time-step method during local nighttime.

4.2 Gradient distribution modeling as a function of local time

As mentioned earlier, severe gradients which occur during local nighttime (18:00 – 06:00 LT) can be produced by EPBs or related instability phenomena. “Non-worst-case gradients” address the gradients which are not extreme but still unusual enough to be relatively rare compared to nominal nighttime gradients. Hence, we selected 34 days from the Phase One study subset of “scintillating days” that we processed to develop the Brazilian threat model [2]. The days in this data subset were selected based on the depth of amplitude scintillation and number of satellites affected by scintillation at different locations within Brazil [2]. Using the station pair method, no gradient larger than 400 mm/km was observed on the selected days. Dual-frequency code and carrier-phase GPS data (L1 at 1575.42 MHz and L2 at 1227.60 MHz) for 24-hour periods from over 180 Brazilian reference stations on the selected days were used as inputs into the Long-Term Ionospheric Anomaly Monitor (LTIAM) software package [13]. Precise ionospheric delays generated by the LTIAM were used to compute the gradients.

As an alternative to the station-pair method, the time-step method estimates ionospheric gradients by dividing a delay difference between one epoch $I(t_1)$ and a later epoch $I(t_2)$ by its IPP travel distance from t_1 and t_2 [14].

By adjusting the appropriate time interval, this technique enables us to reflect and measure the desired spatial scale of ionospheric gradients. Furthermore, this method is not highly sensitive to measurement errors caused from receiver and satellite bias estimation, and thus it is useful for automated bulk ionospheric data processing. One weakness of this method is that the estimated gradient will include temporal as well as spatial changes in ionospheric delay. We tried to minimize this problem by choosing a short interval (one minute) for gradient

computation to reduce temporal gradient errors. At a theoretical height of 350 km, IPPs move with a velocity of about 60 to 600 m/s as a satellite elevation angle varies from 90 to five degrees. The corresponding travel distance of IPP is about 3-36 kilometers in one minute time window. Nevertheless, effects of temporal gradients still remain, making these estimates conservative representations of the spatial gradients that we are interested in.

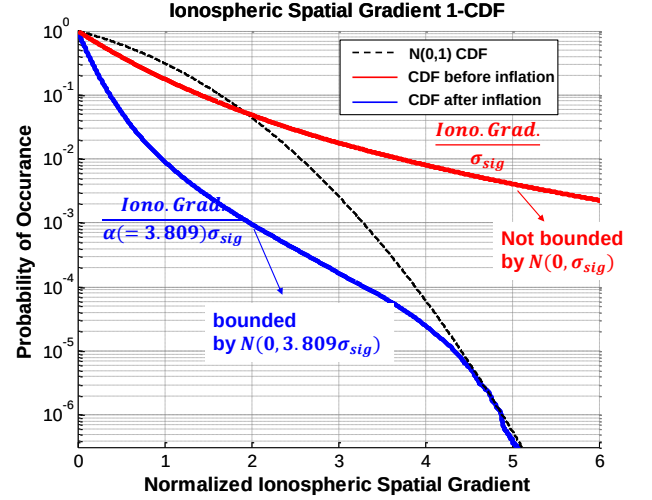


Figure 7. CDF distributions of ionospheric gradients (18:00 – 06:00 LT) before and after applying inflation factor α .

Figure 6 shows gradients computed using the time-step method during local nighttime (18:00-06:00 LT). It is seen that the distribution of the resulting gradients depend highly on local time during the night, with a peak between 21:00 – 23:00 UT. Also, as noted above, the inclusion of temporal gradients in the data causes the largest estimated gradients in Figure 6 to significantly exceed 500 mm/km, despite these being “non-worst-case” estimates.

Using the data from Figure 6, we normalized the estimated ionospheric gradients by calculating sample means and standard deviations in 2-hour bins of local time and obtained the one-minus Cumulative Distribution Function (1-CDF) (combined over all bins) denoted by the red solid curve in Figure 7. The normalized gradient distribution has non-Gaussian tails violating the 1-CDF of the standard normal distribution (as denoted by the black dashed curve). Thus, the sample one sigma of normalized gradients (σ_{sig}) should be increased by multiplying by an inflation factor (α) to cover the non-Gaussian tails of the actual distribution. In this paper, the inflation factor sufficient to bound the combined normalized distribution is determined to be 3.809. The inflated 1-CDF distribution is shown as blue solid curve in Figure 7.

The σ_{sig} overbound is computed as $|\mu_{sig}| + \alpha \sigma_{sig}$ for each 2-hour local time bin [14]. The estimated σ_{sig} overbounds

for each bin are shown in Figure 8 as the red asterisk curve. Using these statistics with local time dependence, we randomly sample anomalous gradients for non-worst-case satellites from a Normal ($0, \sigma_{sig}$ overbound) distribution.

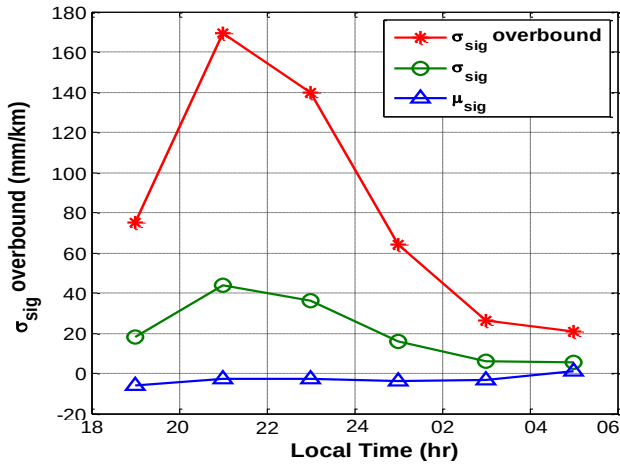


Figure 8. σ_{sig} overbounds during local nighttime (18:00 – 06:00 LT).

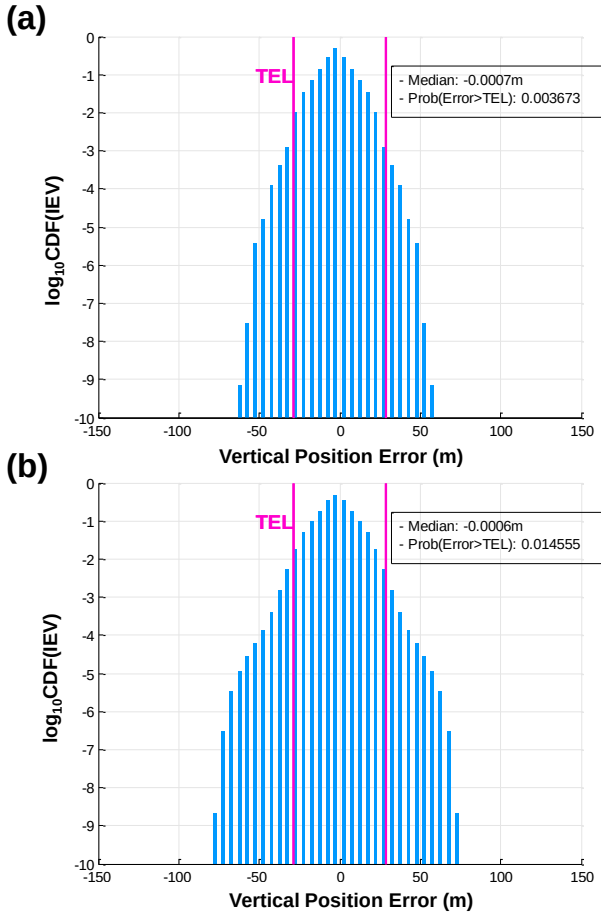


Figure 9. Folded CDF distribution of IEVs from 18:00 to 06:00 LT for (a) the worst one-SV impact case and for (b) the worst two-SV impact case. The vertical magenta lines indicate TEL of ± 28.78 m at CAT-I decision height.

4.3 Real-time threat mitigation using Monte Carlo simulation

Figure 9 shows the folded CDF of ionosphere-induced vertical errors (or IEV) at a DH from the GBAS ground stations resulting from Monte Carlo simulation. The magenta lines indicate the 28.78-meter TEL at the 200-ft CAT-I DH. The probability of errors exceeding TEL for the worst one and two SV impact cases are approximately 0.0037 and 0.015 respectively. The results imply that if we assume a total allowed (sub-allocated) integrity risk of 10^{-8} , the prior probability would need to be no greater than 10^{-6} or 10^{-7} . Unfortunately, while extreme EPB events in Brazil are uncommon, they are significantly more likely than this. Therefore, the Monte Carlo method also needs an additional means of mitigation.

Figure 10 shows the overview of a proposed mitigation scheme for the Monte-Carlo approach that is based on the sigma parameter inflation used in PDGS. For each epoch, we inflate VPL_{H0} and VPL_{eph} of all possible subset geometries and then screen out the geometries whose inflated VPL_{H0} or VPL_{eph} exceed VAL. After that, we conduct Monte Carlo simulation and compute the probability of potentially unsafe errors with the remaining subset geometries. If the computed probability is larger than P_{PL} , which is an allowed probability of errors exceeding the protection level, we increase the sigma inflation factors and repeat the process. This recursive process is iterated until the unsafe error probability becomes less than P_{PL} . Note that P_{PL} of 10^{-5} is derived from the previously sub-allocated integrity risk of 10^{-8} after assuming a prior probability of 10^{-3} for extreme ionospheric gradients. Note that the proposed method is for the real-time use, but Monte-Carlo simulations are conducted offline to determine the required inflation factors in this study.

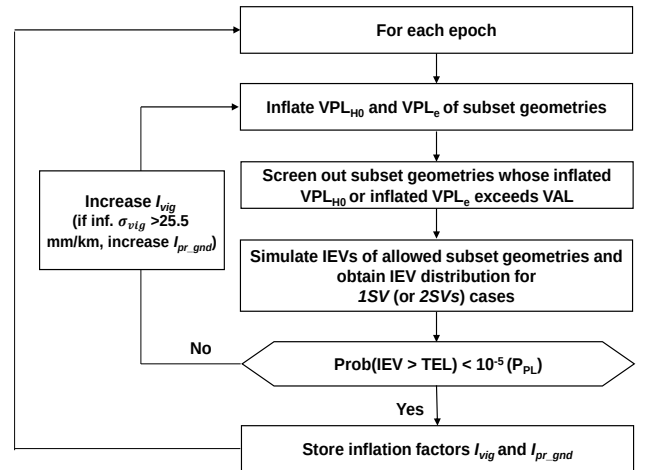


Figure 10. Real-time threat mitigation concept using Monte Carlo simulation.

4.4 Results from EPB impact assessment

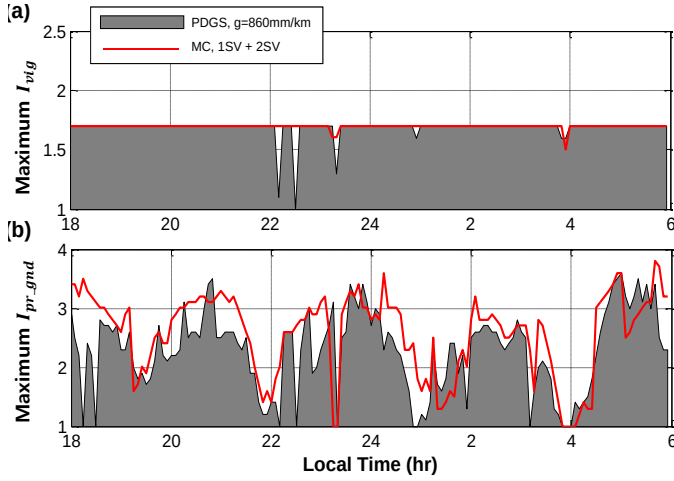


Figure 11. Maximum inflation factors for (a) σ_{vig} and (b) σ_{pr_gnd} for the worst two-satellite impact case from 18:00 to 06:00 LT.

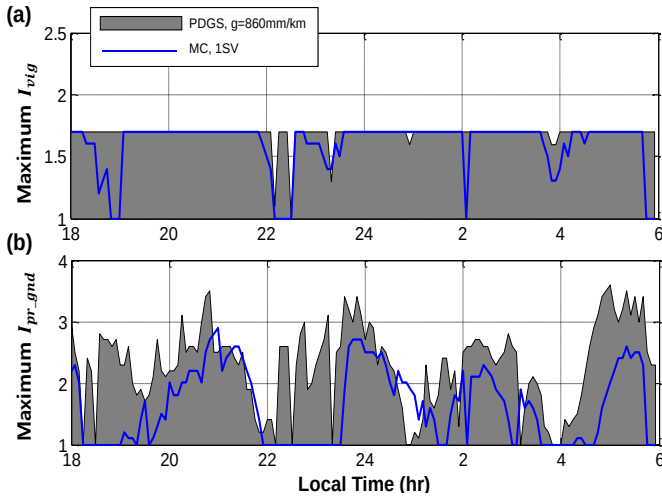


Figure 12. Maximum inflation factors for (a) σ_{vig} and (b) σ_{pr_gnd} for the worst one-satellite impact case from 18:00 to 06:00 LT.

Using the method of Figure 10, scheme, at each epoch, we generate maximum inflation factors for the worst two-satellite impact case (see the red curve in Figure 11), as we did previously for PDGS in Section 3. The inflation factors calculated from PDGS with a maximum gradient of 860 mm/km are denoted in grey in Figure 11. The results show that the Monte Carlo simulation requires higher inflation factors than PDGS almost all the time. This might be surprising, since the Monte Carlo approach is designed to randomize ionospheric scenarios and avoid simulating worst-case conditions at all times. It occurs because the Monte Carlo simulation combines worst two-satellite impact cases with the non-worst-case gradients that were applied to the remaining satellites. Considering that PDGS assumes worst-case impacts on two satellites and no gradient on other satellites, the Monte Carlo

analysis of dual-satellite impacts generates larger errors than PDGS.

Thus, the single worst-gradient-impact case (combined with non-worst-case gradients on the other satellites) is set as the baseline case for the Monte Carlo approach and is compared to the results from the earlier PDGS simulation. The benefit of taking credit for a prior probability of 10^{-3} can be seen in Figure 12. The inflation factors of σ_{vig} and σ_{pr_gnd} from the Monte Carlo method (denoted by the blue curve) are less than those from PDGS most of the time.

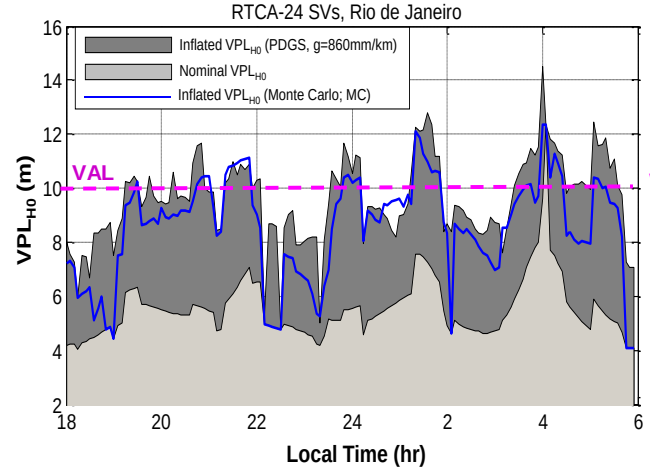


Figure 13. Inflated VPL_{H0} from PDGS (dark grey area) and the Monte Carlo method (blue solid curve), nominal VPL_{H0} (grey area), and VAL (magenta line) at the CAT-I DH. P_{PL} is set to be 10^{-5} .

Table 1. Percentage of time when VPL_{H0} from the Monte Carlo method (MC) is below VPL_{H0} from the PDGS, and Mean VPL difference. P_{PL} is set to be 10^{-5} .

Percentage of time when MC $VPL_{H0} < PDGS$ VPL_{H0}	Mean VPL difference (PDGS $VPL_{H0} - MC$ VPL_{H0})
86.11 %	1.02 m

Figure 13 shows inflated VPL_{H0} values from the two different methods for a CAT I DH located 4 km from the GBAS ground station. The lowered inflation factors for the Monte Carlo method benefit the system availability. Compared to the result from PDGS, the system availability during nighttime increases from 58.33% to 75.00%. To take a closer look, we define two quantities as shown in Table 1. One is the percentage of time when VPL_{H0} from the Monte Carlo simulation is below VPL_{H0} from PDGS, which is 86.11%. The other is the mean VPL difference between PDGS and the Monte Carlo method, and the magnitude of inflated VPL_{H0} from the Monte Carlo method is lower by about 1 m on average.

We examined another possibility in which extreme EPBs are assumed to be rarer events, meaning that their prior probability can be reduced to 10^{-5} , which would raise P_{PL} to 10^{-3} . With this probability, VPL can be lowered further,

and the resulting availability increases to 90%, as shown in Figure 14. Furthermore, as shown in Table 2, the inflated VPL_{H0} values from the Monte Carlo method are almost always lower than those from PDGS, and its magnitude decreases by 2.31 meters on average.

From these results, it is evident that system availability in the presence of extreme EPB-driven gradients is highly sensitive to their prior probability and the resulting P_{PL} . Thus, it is important to estimate and justify these probabilities before implementing Monte Carlo-based ionospheric threat mitigation.

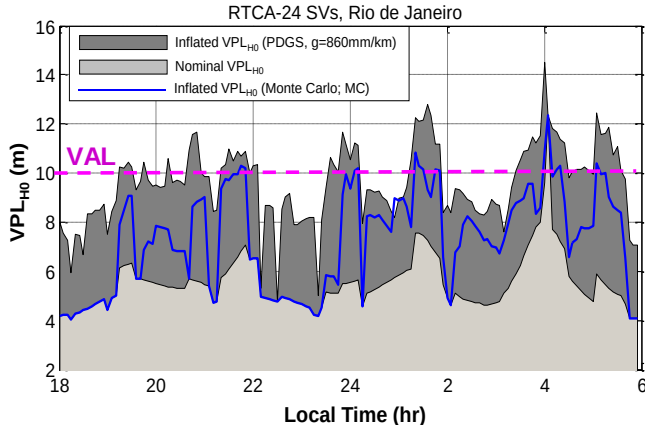


Figure 14. Inflated VPL_{H0} from PDGS (dark grey area) and the Monte Carlo method (blue solid curve), nominal VPL_{H0} (grey area), and VAL (magenta line) at CAT-I DH. P_{PL} is set to be 10^{-3} .

Table 2. Percentage of time when VPL_{H0} from the Monte Carlo method (MC) is below VPL_{H0} from the PDGS, and Mean VPL difference. P_{PL} is set to be 10^{-3} .

Percentage of time when MC $VPL_{H0} < PDGS$ VPL_{H0}	Mean VPL difference (PDGS $VPL_{H0} - MC$ VPL_{H0})
99.31 %	2.31 m

5.0. CONCLUSION

In this paper, we assessed EPB-induced gradients on CAT-I GBAS in the Brazilian region based on the preliminary Brazilian ionospheric threat model derived in previous research [2]. We used two different approaches for this assessment. As an input to the EPB impact simulations, we modeled undetected differential range errors that can be experienced by hypothetical GBAS users under anomalous ionospheric conditions. Mathematical models of the CCD monitor and the carrier smoothing filter were derived using Laplace transforms. An exhaustive simulation algorithm enabled us to estimate the maximum differential range errors over all possible combinations of EPB threat model parameter within the GBAS service volume. Based on these results, we established a simplified analytical model of ionosphere-induced differential range error as a function

of gradient magnitude and a relative speed between the EPB front and the affected satellite IPP. This model enables us to conservatively approximate all possible range errors under worst-case ionospheric impact scenarios.

The first approach was based on the existing PDGS algorithm used in CONUS. While the parameters of the preliminary EPB threat model were incorporated in the simulation, we generated sets of worst-case geometries among EPB fronts and satellite IPPs. These scenarios were organized in a similar manner to that of the CONUS threat model. However, the approach direction and the speed range of EPBs were more constraining on the selection of pairs of impacted satellites compared to the corresponding parameters of the CONUS threat model. Based on these EPB impact scenarios, the performance of GBAS in the Brazilian region under worst-case ionospheric gradient conditions was analyzed. The results showed that, with a maximum gradient greater than about 600 mm/km, it is difficult to meet the CAT-I GBAS availability requirement using this mitigation approach.

In addition, we employed stochastic analysis based on Monte Carlo simulation to evaluate the possibility of potentially unsafe ionospheric errors under the presence of EPBs. This approach assumed that satellites in each subset geometry at each simulated time epoch are affected by the same EPB event, which is specified by randomly sampled threat model parameters. It was further assumed that either one or two satellite IPPs are affected by the largest gradient of 860 mm/km, while the remaining IPPs are affected by anomalous but non-worst-case gradients. These non-worst-case gradients are modeled from data using a bounding Gaussian distribution as a function of local time (nighttime only). Based on the Monte Carlo simulation, we proposed a new mitigation scheme incorporating a prior probability of extreme ionospheric events. For the one-satellite-impact case, the resulting inflation was less conservative than the worst-case approach (PDGS), leading to significantly less availability loss.

We now have knowledge of ionosphere-induced position errors of GBAS users in the presence of EPBs in the Brazilian region. Based on both the worst-case and the Monte Carlo results, future work should consider refinements to the integrity parameter inflation method or some other screening technique beyond that used in the mid-latitude regions. We should obtain help from the additional monitoring and geometry screening included in GAST-D (CAT II/III) GBAS, which is expected to be in service by around 2020. Improved monitoring performance would reduce undetected ionospheric errors and thus improve system availability.

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